

20. (1.5 points)

An insurance company does not have a reinsurance program in place. Based on the latest Enterprise Risk Management review, management wants to reduce its insurance hazard risk by purchasing reinsurance and is reviewing two reinsurance programs.

An actuary estimates the following information based on the modeled probability distribution of the underwriting results:

	Current	Option 1	Option 2
Direct Premium (\$000)	1,000	1,000	1,000
Direct Loss (\$000)	600	600	600
Net Expenses (\$000)	200	200	200
Expected Ceded Premium (\$000)	0	220	330

Expected Net Combined Ratio	80.0%	83.0%	86.0%
1-in-100 TV@R Net Loss (\$000)	2,000	1,500	1,100

- The company uses 1-in-100 TV@R to set its required capital.

Calculate the insurance company's marginal ROEs for reinsurance Option 1 and Option 2.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 20

TOTAL POINT VALUE: 1.5

LEARNING OBJECTIVE: C2

SAMPLE ANSWERS

Sample 1

	<u>Current</u>	<u>Option 1</u>	<u>Option 2</u>
Capital held	2,000	1,500	1,100
Δ Capital	-	-500	-900
Net Prem <i>=Direct-Ceded</i>	1,000	780	670
Net Profit <i>=(Prem)(1-CR)</i>	200	132.6	93.8
Δ Profit	-	-67.4	-106.2
Marginal ROE <i>= ΔProfit / ΔCapital</i>		13.48%	11.8%

Sample 2

$$\text{Marginal ROE} = \frac{\text{Net Cost of Reinsurance}}{\text{Capital Freed}}$$

$$\text{Capital Freed}_1 = 2,000 - 1,500 = 500$$

$$\text{Capital Freed}_2 = 2,000 - 1,100 = 900$$

$$\text{Net Cost}_1 = \text{Ceded Prem} - \text{Recoveries} = 220 - 152.6 = 67.4$$

$$CR_1 = 0.83 = \frac{\text{Net Expense}}{\text{Net Prem}} + \frac{\text{Net loss}}{\text{Net Prem}} = \frac{200 + \text{Net loss}}{1,000 - 220}$$

$$\text{Net Loss}_1 = 447.4$$

$$\text{Ceded Loss}_1 = 600 - 447.4 = 152.6$$

$$CR_2 = 0.86 = \frac{\text{Net Expense}}{\text{Net Prem}} + \frac{\text{Net loss}}{\text{Net Prem}} = \frac{200 + \text{Net loss}}{1,000 - 330}$$

$$\text{Net Loss}_2 = 376.2$$

$$\text{Ceded Loss}_2 = 600 - 376.2 = 223.8$$

$$\text{ROE}_1 = 67.4 / 500 = 0.1348$$

$$\text{ROE}_2 = 106.2 / 900 = 0.118$$

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

EXAMINER'S REPORT

Candidates were expected to calculate the marginal net cost of each of the two reinsurance programs by using the change in Net Underwriting Income from the Current, and using the difference between Ceded Losses and Ceded Premiums.

Candidates were then expected to ratio this cost to the marginal change in required capital, as disclosed by the 1-in-100 TV@R value, to arrive at a marginal ROE for each reinsurance program option.

Common mistakes included:

- Incorrect calculations of Net Loss or Net Underwriting Income (e.g. applying the Net Combined Ratio to gross premium instead of net premium, or assuming the expense ratio was fixed at 20% for each option).
- Calculations of Reinsurance program costs (e.g. using the Expected Ceded Premium as the "cost" of the reinsurance program, or calculating marginal cost as $(\text{Net Combined Ratio} - 80\%) \times \text{Gross Premium}$).
- Calculating the ROE of each option as Net UW Income divided by Required Capital and then calculating the marginal ROE as the difference in these ROEs.